

This file includes the: (1) Source Code and (2) Program Output.

Source Code

.java file link: [ExpenseTracker_Karim](#)

```
import java.util.Scanner;

public class ExpenseTracker_Karim {

    // output: program title (void)
    public static void title(){
        System.out.println("=== Personal Expense Tracker ===");
    }

    // process: calculation of total expenses (non-void)
    public static double totalExpenses(double food, double transpo, double acadNeeds,
double others){
        return food + transpo + acadNeeds + others;
    }

    // process: checking of budget status (non-void)
    public static String budgetCheck(double totalExpenses, double budget){
        if(totalExpenses > budget){
            return "Oh no! You have exceeded your budget!";
        }else{
            return "Good job! You're still within your budget!";
        }
    }

    // output method: displays total and budget status (void)
    public static void result(String name, double totalExpenses, double budget, String
budgetStatus){
        double remaining = budget - totalExpenses;

        System.out.println("\n===== " + name + " Expense Summary =====");
    }
}
```

```
System.out.println("Total Expenses: " + totalExpenses);
System.out.println("Budget: " + budget);

if(remaining < 0){
    System.out.println("Budget Overrun: " + (totalExpenses - budget));
    remaining = 0;
}

System.out.println("Remaining: " + remaining);
System.out.println("Status: " + budgetStatus);
}

public static void main(String[] args){

    Scanner input = new Scanner(System.in);

    double food, transpo, acadNeeds, others, budget;
    String name;

    // user input part
    title(); // prints title method

    while(true){
        System.out.print("Enter your name: ");
        name = input.nextLine();
        // validate name: letters only
        if(name.matches("[a-zA-Z ]+")){
            break;
        }else{
            System.out.println("Invalid name input");
        }
    }

    while(true){
        System.out.print("Enter your Food expense: ");
```

```
// validate amount: double only
if(input.hasNextDouble()){
    food = input.nextDouble();
    break;
}else{
    System.out.println("Invalid amount input");
    input.next();
}
}

while(true){
    System.out.print("Enter your Transportation expense: ");
    // validate amount: double only
    if(input.hasNextDouble()){
        transpo = input.nextDouble();
        break;
    }else{
        System.out.println("Invalid amount input");
        input.next();
    }
}

while(true){
    System.out.print("Enter your Academic Needs expense: ");
    // validate amount: double only
    if(input.hasNextDouble()){
        acadNeeds = input.nextDouble();
        break;
    }else{
        System.out.println("Invalid amount input");
        input.next();
    }
}

while(true){
```

```
        System.out.print("Enter your Other expenses: ");
        // validate amount: double only
        if(input.hasNextDouble()){
            others = input.nextDouble();
            break;
        }else{
            System.out.println("Invalid amount input");
            input.next();
        }
    }

    while(true){
        System.out.print("Enter your Budget: ");
        // validate amount: double only
        if(input.hasNextDouble()){
            budget = input.nextDouble();
            break;
        }else{
            System.out.println("Invalid amount input");
            input.next();
        }
    }

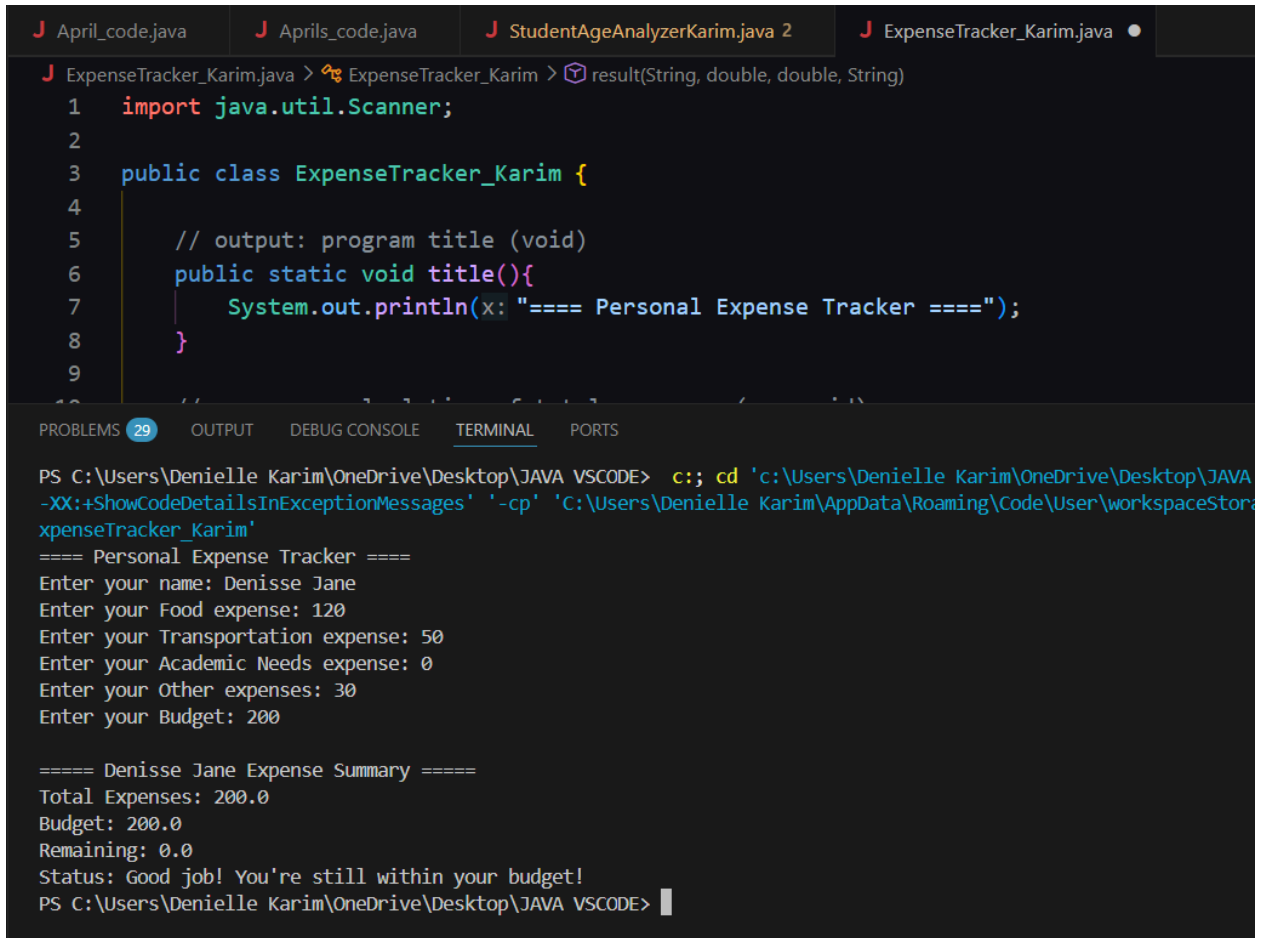
    // process: calculation
    double totalExpenses = totalExpenses(food, transpo, acadNeeds, others);
    String budgetStatus = budgetCheck(totalExpenses, budget);

    // output: call method
    result(name, totalExpenses, budget, budgetStatus);

    input.close();
}
}
```

Program Output

Scenario 1: User expends within budget and no input errors.



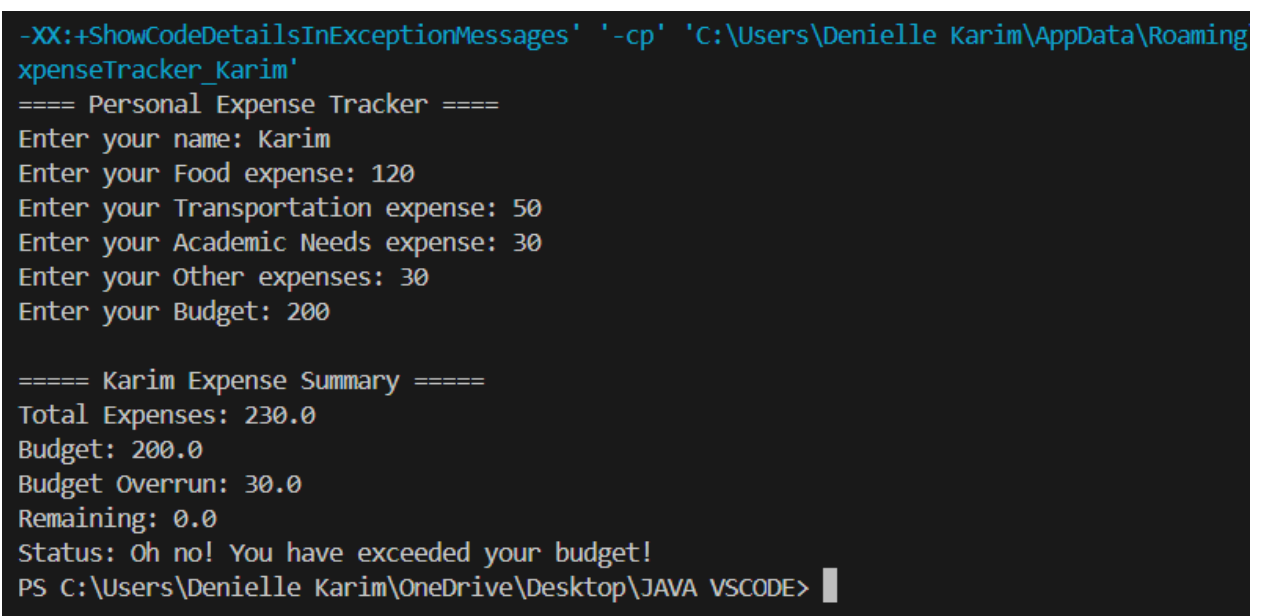
```
J April_code.java J Aprils_code.java J StudentAgeAnalyzerKarim.java 2 J ExpenseTracker_Karim.java •
J ExpenseTracker_Karim.java > ExpenseTracker_Karim > result(String, double, double, String)
1  import java.util.Scanner;
2
3  public class ExpenseTracker_Karim {
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5      // output: program title (void)
6      public static void title(){
7          System.out.println(x: "==== Personal Expense Tracker ====");
8      }
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PROBLEMS 29 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Denielle Karim\OneDrive\Desktop\JAVA VSCODE> c::; cd 'c:\Users\Denielle Karim\OneDrive\Desktop\JAVA
-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Denielle Karim\AppData\Roaming\Code\User\workspaceStor
xpenseTracker_Karim'
==== Personal Expense Tracker ====
Enter your name: Denisse Jane
Enter your Food expense: 120
Enter your Transportation expense: 50
Enter your Academic Needs expense: 0
Enter your Other expenses: 30
Enter your Budget: 200

===== Denisse Jane Expense Summary =====
Total Expenses: 200.0
Budget: 200.0
Remaining: 0.0
Status: Good job! You're still within your budget!
PS C:\Users\Denielle Karim\OneDrive\Desktop\JAVA VSCODE>
```

Scenario 2: User expends over budget (added an Overrun result) and no input errors.



```
-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Denielle Karim\AppData\Roaming
xpenseTracker_Karim'
==== Personal Expense Tracker ====
Enter your name: Karim
Enter your Food expense: 120
Enter your Transportation expense: 50
Enter your Academic Needs expense: 30
Enter your Other expenses: 30
Enter your Budget: 200

===== Karim Expense Summary =====
Total Expenses: 230.0
Budget: 200.0
Budget Overrun: 30.0
Remaining: 0.0
Status: Oh no! You have exceeded your budget!
PS C:\Users\Denielle Karim\OneDrive\Desktop\JAVA VSCODE>
```

Scenario 3: Program loops back to where the user input an error, proceeds only when satisfied.

- The Expense Summary will only print with complete valid inputs.
- Name only accepts letters a-z, A-Z
- Expenses and budget only accepts double

```
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le Karim\AppData\Roaming\Code\User\workspaceSto
==== Personal Expense Tracker ====
Enter your name: 676767
Invalid name input
Enter your name: Nissy
Enter your Food expense: wala
Invalid amount input
Enter your Food expense: 0
Enter your Transportation expense: 10
Enter your Academic Needs expense: █
```